University of Prishtina “Hasan PRISHTINA”

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Week II – Data Gathering

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Data Gathering Approaches

- **API-s**:
  - **A**pplication **P**rogramming **I**nterface, is a set of rules and protocols that allows different software applications to communicate with each other;

- **Crowdsourcing**:
  - Crowdsourcing refers to the process of collecting data, information, or services from a large group of people, typically via the internet.

Data Gathering Approaches

- **Regular Expressions**:
  - A regular expression defines a search pattern, allowing you to match and extract specific pieces of text from a larger body of data;
- **Crawling:**
  - A crawler, often referred to as a web crawler, web spider, or web robot, is a computer program or script that systematically navigates the internet by visiting websites and web pages, collecting information from them.
  - What is difference between web **crawler** and **scraper**?
    - A **crawler** is a program or bot designed to **browse the web** automatically and index content from different websites.
    - A **scraper** is designed to **extract specific** data from web pages.

Data Gathering Approaches

- What is web **Crawler**, **Spider** and **Robot**?
  - A **Web crawler** is a program or bot designed to browse the web automatically and index content from different websites.
  - A **web spider** is an automated program or script that systematically browses the web, indexing content from websites.
  - A **web robot** (also known as a bot or internet bot) is a software application that runs automated tasks over the internet. These tasks are typically repetitive and can be executed much faster than by a human.

DATA GATHERING: API-S

- Ready-made targeted tools to interface – get fed by the target source via a graphical or command-line tool;
  - Example: https://exportcomments.com/

- Common features:
  - API Key which refers to the receiving page of the data;
  - Format of data in xls, csv, JSON, XML.

Data Gathering: Weather API Example

- https://www.weatherapi.com/
  - Air pollution, important parameters: air pressure, air humidity, compare min and max temp;
  - Proprietary data.
- https://airnow.gov
  - Air quality data;
  - Proprietary data.
- https://openweathermap.org
  - e.g., 5-day weather data every 3 hrs.

Data Gathering: Weather API Example (CONT.)

- ***ASK te dhena ne Kosove:***
  - *Geographical data, air temperature (aggregated data only per month);*

- ***Air quality prej IHMIK***
  - *https://airqualitykosova.rks-gov.net/en/*

Data Gathering: Crowdsourcing

- Engage the certain community group to contribute with data provision:
  - A model in advance to ensure the success in data quality and data completion

- Example: https://exportcomments.com/

Data Gathering: Regular Expressions

- Patterns to semi-automate the data extraction process;

- Example:
  - *PCRE (Perl Compatible Regular Expressions), POSIX (Portable Operating System Interface), re module in Python, Boost.Regex in C++, .NET Regex in C#, Oniguruma in Ruby, Regex package in Java, RegExp object in JavaScript and so on.*

Data Gathering: Crawling

- Crawling vs. Scraping, also terms Wrapper, Spider in use:
  - To automatically collect data from the existing pages on the Web
    - https://developers.google.com/search/docs/advanced/guidelines/how-search-works
    - http://nlp.stanford.edu/IR-book/html/htmledition/web-crawling-and-indexes-1.html
- Types of Crawlers:
  - HTML-aware
  - NLP-based
  - Structure-based
  - Ontology-based
  - Machine / Deep learning approach

Data Gathering: Crawlers

- **XPather** (http://xpather.com/) (see also FireBug, Firepath, Zotero, etc.);
- **XWrap** (http://www.cc.gatech.edu/projects/disl/XWRAPElite/);
- **Scrapy** (https://scrapy.org/);
- **Web-Harvest** (https://sourceforge.net/projects/web-harvest/);
- **Web Data Extractor**;
- **Crawler Workbench**;
- **LAPIS** (https://lapisdata.com/);
- Araneus , Jedi, Road Runner, etc.

Data Gathering: Target Crawlers

- NetPeakSoftware lists target pages and appropriate scrappers;

- Scraping for Amazon recommended APIs:
  - https://www.webscrapingapi.com/ for Amazon;

- Free Google Search Scraper:
  - SerpAPI (https://serpapi.com/).

Data Gathering: Definition

"a legal instrument that specifies a standard set of terms and conditions regarding sharing and re-use of your research data"*

- **Licenses for software:**
  - *MIT, Apache, BSD, GPL, MPL (Mozilla), CeCILL;*
- **Licenses for hardware:**
  - CERN OHL, TAPR OHL (Open Source);
- **Licenses for data?**

* Source: https://www.wur.nl/en/Value-Creation-Cooperation/Collaborating-with-WUR-1/WDCC/DataManagement-WDCC/Finishing/Data-Licences.htm

DATA SHARING OPTIONS

Data Licenses: Creative Commons (CC)

- *The Creative Commons (CC) licenses provide a flexible range of protections and freedoms for creators and their works.*

- *Specifically, CC0 (Creative Commons Zero) and CC BY licenses are widely used, and different versions (1.0, 2.0, 3.0, and 4.0) represent updates to those licenses over time.*

- *Let's break down CC0 1.0, and the differences in versions of the CC BY 4.0 license.*

CC0 1.0 (Creative Commons Zero)

- **Definition: CC0 1.0**
  - *Universal (CC0) is a public domain dedication tool. When a work is released under CC0, the creator waives all copyright and related rights to the extent allowed by law, effectively putting the work in the public domain. Anyone can use, modify, distribute, or build upon the work without needing to ask for permission.*

- **Purpose:**
  - *To make the work free of any copyright restrictions, allowing anyone to use it for any purpose without attribution.*

CC0 1.0

- **Released:** *2002*

- **Definition:**
  - *The first version of the Creative Commons license. It allowed creators to offer their work with the condition that the user gives attribution (credit) to the original creator. This version was an early attempt at balancing copyright protection with the need for free sharing of knowledge.*

- **Limitations:**
  - *This version was more regionally focused and didn't fully address international legal variations in copyright law.*

CC BY 2.0

- **Released:** *2004*

- **Definition:**
  - *Introduced updates to improve the legal code of the license and internationalized it more effectively, allowing it to be adopted in a broader range of jurisdictions.*

- **Changes:**
  - *Made the attribution requirement clearer and improved the license to be more globally applicable.*

CC BY 3.0

- **Released:** *2007*

- **Definition:**
  - *Continued to improve the legal code and further adapted the license for international use.*

- **Notable Changes:**
  - *Moral rights (rights of creators to claim authorship or object to certain uses) were better accounted for;*
  - *The license became more robust and aligned with the increasing use of digital technologies and the internet.*

CC BY 4.0

- **Released:** *2013*
- **Definition**:
  - *The most recent version of the Creative Commons license. It provides global coverage and is designed to be more internationally effective.*
- **Notable Changes:**
  - ***Interoperability:*** *Easier to combine with other licenses, like the GPL (General Public License).*
  - ***Database rights:*** *The license now covers database rights in jurisdictions that recognize them, making it easier to apply CC BY 4.0 to datasets.*
  - ***Improved attribution:*** *More flexible requirements for giving credit, recognizing that in some cases, full attribution may not be possible.*
  - ***Globalization:*** *More emphasis on ensuring the license works across different legal systems.*

Class work

- **Each of you turn around and make a group from 3-4 students and for each of you it will be assign one of these task. Choose any website which share news?**

- **Task 1:**
  - *Crawling only links from a web site;*

- **Task 2:**
  - *Crawling only Titles from a web site;*

- **Task 3:**
  - *Crawling only Content from a web site.*

Questions?

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